

EduMind AI System Architecture

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Team ID xxxxxx
Project Name EduMind AI - Smart AI-Powered Learning Assistant
Maximum Marks 3 Marks

Step 1: Architecture Overview

EduMind AI follows a layered architecture that separates the user interface, backend routing, feature modules and external AI service.

Step 2: Architecture Components

Layer	Component	Responsibility
1. Presentation	HTML / CSS / JavaScript Frontend	Collects user input, triggers feature requests and displays responses.
2. Application	FastAPI Backend	Receives HTTP requests, validates input and routes requests to feature modules.
3. Feature Modules	Q&A, Explanation, Quiz, Notes, Summary, Learning Path	Build feature-specific prompts and manage the corresponding learning workflow.
4. AI Integration	Google Gemini API / Gen AI SDK	Processes prompts and generates AI-powered responses.
5. Configuration	Environment variables / .env	Stores API credentials separately from application source code.

Step 3: Request Flow

User → Frontend → FastAPI Backend → Feature Module → Gemini API → Feature Module → Backend → Frontend → User

Step 4: Architecture Goal

Keep the application modular, maintainable and easy to extend so additional educational features can be added without changing the entire system.